

Lu Yao Chen

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EDUCATION

University of Southern California

Aug 2024 - May 2026

Master's of Science in Quantum Information Science

University of Waterloo

Sep 2019 - Jun 2024

Bachelor's in Applied Science in Computer Engineering

RESEARCH

ML Research Assistant - **Zhao Research Group, USC** - Los Angeles, CA

Feb 2025 – Present

- Decreased data loss and accelerated **Physics-Informed Neural Network** (PINN) training **20x** through parallelized and vectorized computation with optimized data structures
- Improved training stability in multi-objective PDE optimization by applying GradNorm dynamic loss weighting

EXPERIENCE

AI Engineer Intern - **eBay Inc.** - San Jose, CA

May 2025 – Aug 2025

- Automated root cause analysis with an **AI-driven** incident response system, cutting diagnosis time **80%**
- Built a **Model-Context-Protocol (MCP) server** for Kubernetes to retain reasoning context across workflows

Cell Engineer Intern - **Tesla** - Fremont, CA

May 2023 – Dec 2023

- Designed a relational schema and asynchronous cron-driven pipeline to centralize cathode production gauge data, cutting ingestion time **60%** and improving data integrity
- Built a real-time dashboard for gauge and production data to enable root-cause analysis of battery efficiency

Software Engineer Intern - **1Password** - Toronto, ON

Jan 2023 – Apr 2023

- Developed a billing service with Salesforce pub/sub APIs enabling real-time synchronization and reducing errors

Software Engineer Intern - **Fragment Foundries** - Seattle, WA

Jan 2022 – Apr 2022

- Built a double-entry accounting system and dashboard, raising test coverage **25%** and reducing demo defects **30%**

Software Engineer Intern - **Air Matrix** - Toronto, ON

May 2021 – Aug 2021

- Built Go microservices for drone operations, enabling real-time analytics and automated flight management

Software Engineer Intern - **Wind River** - Ottawa, ON

Sep 2020 – Dec 2020

- Automated sensitive log redaction with Python/Bash, boosting debugging info 250% and strengthening security

Software Engineer Intern - **Tucows** - Toronto, ON

Jan 2020 – Apr 2020

- Automated OpenStack deployment with Terraform/SaltStack, boosting efficiency **200%** and reducing errors **30%**

PROJECTS

FAQ Service

Jan 2023

- Developed a tool for deploying answering services streamlining the process to create and manage their own FAQ

GPTDiscord

Mar 2023

- Maintained and contributed to GPTDiscord, a Python-based communication platform integration

SKILLS

Languages: Python, Go, C/C++, TypeScript, SQL, Bash, Terraform, HTML/CSS

Tools: PyTorch, Qiskit, React, GraphQL, Kubernetes, Docker, Postgres, AWS, Airflow, Django